

Project Requirement Document

DOCUMENT NUMBER

PROJECT TITLE: Streamlining Tech for Client Usability (STeCU)
CLIENT: PM Division





Table of Contents

Team Structure	3
Project Overview	4
1. Executive Summary	4
2. Design and Implementation Constraints	4
3. Assumptions and Dependencies	5
Project Requirements	6
4. Interfaces	6
5. Functional Requirements	7
6. Non-Functional Requirements	8
Project Execution	10
7. Project Scope	10
8. Project Deliverables	10
9. Timeframe and Milestones	10
10. Challenges and Open Questions	11



Team Structure

- Project Manager : Bayu Putra Ibana
- Client : Project Manager Division
- Project Officer : Yusuf Imantaka Bastari
- Team
 1. Bayu Putra Ibana
 2. Kartika Adhi N.W
 3. Keyne Elvaretta
 4. Yusuf Imantaka Bastari
 5. Revy Satya Gunawan



Project Overview

1. Executive Summary

The Project Management division in OmahTI faces several challenges related to client handling. One recurring issue is that returning clients often encounter different development teams, as OmahTI's staff is renewed annually. However, a more pressing concern is the lack of new client acquisition. To address this, we focused on creating a feature that would attract and onboard potential clients more effectively.

We drew inspiration from a unique feature commonly found on charity websites, where users can input specific criteria—such as the purpose of the donation, preferred amount, and timeline—and instantly receive a transparent breakdown of how their contributions will be allocated. Adapting this concept, we proposed the development of a rate-card estimator on the OmahTI website. This feature streamlines client registration by guiding users through a series of onboarding questions designed to capture their project needs, timeline, service type, and intended purpose. Based on this input, the system will generate an estimated price range, giving clients a clear expectation of costs before entering direct communication—lowering entry barriers and encouraging more engagement.

2. Design and Implementation Constraints

The implementation of the rate-card estimator and client onboarding feature on OmahTI's website faces several design and execution challenges:

- **Time Constraints:** All development and deployment must align with the internship timeline and live mentoring sessions. Additionally, the availability of developers may vary due to academic schedules.
- **Staff Turnover:** OmahTI updates its staff yearly, resulting in current members not being available for long-term maintenance. Therefore, the system must be simple, well-documented, and easily handover-able.
- **Client Use-Case Diversity:** The solution must handle very different client requests (event websites with low budgets and e-commerce clients with higher complexity). This results in large price variations and complex form logic.
- **Inconsistent Client Requests:** Clients often change feature requests during or after project completion, which can lead to inefficiencies and scope creep. The system should clearly define requirements early on and include scope confirmation steps.



3. Assumptions and Dependencies

To successfully execute the project, several assumptions and dependencies are acknowledged:

- Assumptions:
 1. Clients are willing to fill in detailed onboarding questions to help OmahTI understand their needs.
 2. Clients will have varying knowledge levels about tech, so the UI must be simple, explanatory, and supportive.
 3. Price estimation is not final and it is assumed that clients understand it is negotiable and subject to validation by EA (External Affairs) OmahTI.
 4. One person from EA can be assigned to respond to submissions and handle negotiation/pop-up inquiries.
- Dependencies:
 1. Project depends on availability and responsiveness of EA division to respond to incoming submissions.
 2. Accurate pricing depends on OmahTI's internal documentation of past projects and their associated cost ranges.
 3. Google Sheets API or similar service is needed for saving client submissions in a manageable, shared format.
 4. Continuous communication with PM and design mentors is crucial for validating UX decisions and ensuring smooth implementation.
 5. The project's success also relies on proactive communication with potential clients to avoid miscommunications or unexpected feature requests.



Project Requirements

4. Interfaces

4.1. User Interfaces

The website will have a simple and user-friendly interface that focuses on ease of use and clarity. Users will primarily interact with two key components:

1. **Input Form**

A clean and structured form where users can input necessary data related to a task or project to calculate the appropriate rate.

2. **Result Display Page**

After submitting the form, users will be shown the calculated rate card result clearly and immediately.

The design style will follow a **minimalist and human-centered** approach, making sure the layout is not overwhelming and easy to navigate. The UI will also follow **OTI's existing branding guidelines**, including:

- Official colors
- Typography
- Logo usage

The interface will be responsive (mobile- and desktop-friendly) and designed to ensure readability and comfort, even for users with basic digital skills.

4.2. Software Interfaces / Tech Stack

- **UI/UX :**
 - Figma
- **Front-End:**
 - [React.js](#)
 - TailwindCSS
- **Back-End:**
 - [Node.js](#) , Express
 - Axios
- **Other:**
 - Gemini API
 - Google Sheets API



5. Functional Requirements

5.1. User Managements

- FR-1: The system must allow users to access the rate estimator without login (public access).
- FR-2: Admin (EA/PM) must be able to access client submissions through a secure interface (e.g., Google Sheets or dashboard).

5.2. Client Submission Flow

- FR-3: The system must allow users to select a service type (e.g., web dev, mobile app, game dev).
- FR-4: The system must allow users to input the project timeline (start date or deadline).
- FR-5: The system must allow users to specify project purpose/use-case (e.g., event, organization, business).
- FR-6: The system must allow users to input expected budget or preferred price range.
- FR-7: The system must allow users to input additional technical preferences (e.g., desired tech stack).
- FR-8: The system must allow users to submit additional questions or clarifications via an optional text box.
- FR-9: The system must allow users to submit the form, which will be stored and sent to the EA team.

5.3. Estimation Logic

- FR-10: The system must process user inputs and generate a price range estimate using the Gemini API.
- FR-11: The system must display the estimated price along with a disclaimer that the cost is negotiable.
- FR-12: The system must save submission data to Google Sheets via API for team reference and follow-up.

5.4. Client Handoff & Follow-up

- FR-13: The system must display a popup message or confirmation screen after form submission, informing users to await a response from the OmahTI EA team.
- FR-14: The system must show contact information (WhatsApp or email) for clients who want to negotiate or ask questions directly.
- FR-15: The system must include the estimated price range generated from the Gemini API in the confirmation screen, so the client has an initial idea of project cost.



- FR-16: The system must store the estimation result alongside the client's form data in Google Sheets for easy access and follow-up by the EA division.
- FR-17: The system must notify the EA team when a new submission is received (e.g., through Google Sheets update, email, or dashboard alert).

5.5. **Navigation and Landing Page**

- FR-18: The system must display a “Register as Client” button in the website's main navigation bar, ensuring visibility from any page.
- FR-19: The landing page must serve as a promotional space for OmahTI, highlighting past projects, available services, testimonials, and why potential clients should trust and collaborate with OmahTI.

6. **Non-Functional Requirements**

- **Web Performance:**
The website should load quickly to ensure a smooth user experience. With fast loading times, it helps keep visitors engaged.
- **Availability:**
Once the website is deployed, it should be reliably accessible by users. It should almost always be online, except for rare maintenance periods or unexpected issues.
- **Security:**
The website must implement basic security measures, such as input validation to prevent malicious data from being submitted. It should also use HTTPS to secure data transmission between the server and the user's browser.
- **Usability:**
The website must have a simple, clear, and easy to use design. Users should be able to navigate and use the website easily.
- **Maintainability:**
The website's source code must be written in a clean, organized way, use proper formatting, comments and logical file structure. This will make it easier for other developers to understand, fix, or update the website without any issues.



- **LLM Computation Time & Accuracy**

The integrated Large Language Model (LLM) must provide prediction results efficiently and accurately. The LLM's price prediction accuracy for the rate card should be accurate based on the data of OmahTI's previous projects as well as client interviews on IT project pricing.

Regular testing and monitoring should be performed to ensure that both the response time and prediction quality remain consistent.



Project Execution

7. Project Scope

We will design a clean and responsive website layout featuring two main pages: the **Homepage** and the **Register as Client** page. The **Homepage** will offer a concise introduction to OmahTI, an overview of our ongoing and completed projects, as well as key highlights that showcase our achievements and skills. **The Register as Client** page will provide a form for potential clients to submit their contact information along with a rate card calculation based on their project request. Along with that, we will add contact information to our team for further discussion with the client. To enhance the overall user experience, the website will include smooth animations and seamless navigation between pages. Finally, the website will be deployed to a live hosting environment.

8. Project Deliverables

As a result of the scope mentioned above, OmahTI will deliver:

- Fully functional web-based rate card calculator,
- All source code files,
- Design assets (mockups, wireframes),
- A link to the live deployed website,
- Deployment to production server,
- Final presentation.

9. Timeframe and Milestones

Milestone	Date	Description
Ideation Process	16 March 2025	Initial brainstorming and concept development.
Live Mentoring #1 (PM,UI/UX)	25 March 2025	Online session with mentors to refine ideas and strategies.
Interview Session with Potential User	27 March 2025	User interviews to validate the problem space and needs.
Design Sprint	30 March 2025	Collaborative design sprint to define user journeys and create prototypes.



Live Mentoring #2 (PM,BM)	26 April 2025	Online session with mentors to refine ideas and strategies.
Testing	27–29 April 2025	Usability and functionality testing.
Deployment	30 April 2025	Final implementation and launch of the product.
Presentation	2 Mei 2025	Presentation of the project

10. Challenges and Open Questions

- **Client Communication & Scope Creep:**
Mid-project requests for scope changes and unclear requirements from clients may lead to repeated rework and changes in the project timeline.
- **Customized Pricing Model:**
Each client's budget and willingness to pay may vary significantly, making it difficult to define a general pricing structure for each project that we are offering.
- **Marketing & Client Adoption:**
Convincing potential clients or users of the value of OmahTI's services and driving the use of a newly made/implemented feature.
- **Documentation & Knowledge Sharing:**
Assets that are needed such as project portfolio data, organizational achievements, and divisional structures of OmahTI in the previous years are not fully documented or easily accessible to make our price estimation more accurate.
- **Timeline Constraint:**
All deliverables must be finalized within a one-month window.



Open Questions:

1. How can we formalize mid- project change requests to minimize misunderstandings and challenges?
2. What documentation templates should be introduced to ensure all parties agree before work proceeds?
3. What “hook” or marketing narrative will resonate most with our target segments?



Gedung Fakultas MIPA UGM, Sekip 
Utara, Bulaksumur, Sinduandi, Mlati,
Sleman, D.I Yogyakarta

omahti.mipa@ugm.ac.id 

Visit us on
omahti.web.id